

Project Report

Retail Sales Analysis

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1. Project Overview

This project presents a comprehensive Retail Sales and Inventory Intelligence System built for a bicycle retail company operating across three stores in the United States. The goal is to analyze end-to-end retail operations — from order processing and inventory management to staff performance and customer behavior — using a structured, multi-tool analytics pipeline.

Problem Statement The retail company operates multiple stores and needs data-driven insights to improve sales strategies, identify top-selling products, understand customer preferences, optimize inventory, and monitor staff performance across regions.

Project Objectives

- Design and implement a relational database for retail analytics
- Clean and preprocess raw data using Excel
- Import structured data into MySQL and perform integrity checks
- Write SQL queries for multi-dimensional business analysis
- Create reusable SQL Views for Tableau integration
- Build interactive Tableau dashboards covering Sales, Inventory, and Customer & Staff performance
- Generate actionable business insights and recommendations

Deliverables

- Cleaned Excel workbook with 9 tables
- MySQL database with schema, queries, and views
- 3 interactive Tableau dashboards
- Final project report (this document)
- 8-minute recorded walkthrough video

2. Dataset Description

The dataset is a relational retail schema comprising 9 tables spanning two business domains — Sales and Production. It contains 5,000+ transactions across orders, products, customers, stores, staff, and inventory.

Dataset Summary

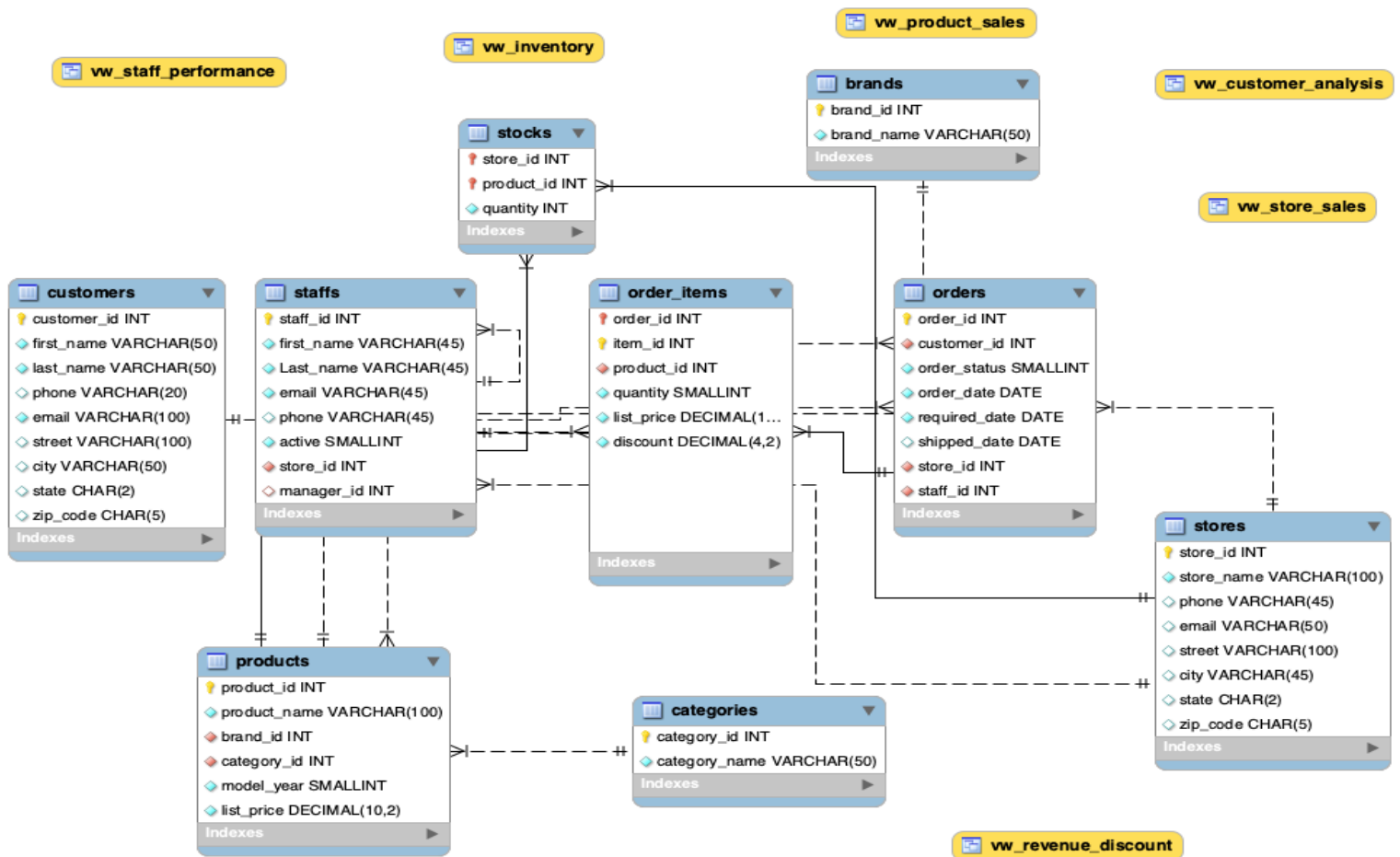
Table	Row Count	Description
brands	9	Bicycle brand names
categories	7	Product category classifications
customers	1,445	Customer demographics and contact details
Stores	3	Store locations across NY, CA, TX
staffs	10	Staff details with manager hierarchy
products	321	Product catalogue with brand, category, price
orders	1,615	Customer orders with status and dates
order_items	4,722	Line items per order with quantity and discount
stocks	939	Current inventory levels per store per product

Domain Structure

- Sales Domain: orders, order_items, customers, staffs, stores
- Production Domain: products, stocks, brands, categories

3. Database Design

Entity Relationship Overview The database follows a normalized relational design with clearly defined primary keys, foreign keys, and constraints. The schema was designed in MySQL Workbench using the EER Model Designer and forward engineered into the live database.



Primary Keys & Foreign Keys

Table	Primary Key	Foreign Keys	Notes
brands	brand_id	-	Parent table
categories	category_id	-	Parent table
customers	customer_id	-	Parent table
Stores	store_id	-	Parent table
staffs	staff_id	store_id, manager_id (self)	Self-referencing FK
products	product_id	brand_id, category_id	-
orders	order_id	brand_id, category_id	-
order_items	(order_id, item_id)	order_id, product_id	Composite PK
stocks	(store_id, product_id)	store_id, product_id	Composite PK

Relationship Integrity Check All foreign key relationships were verified using SQL queries. Every relationship returned 0 orphaned rows, confirming complete referential integrity across all 9 tables.

relationship	orphaned_rows
products → brands	0
products → categories	0
orders → customers	0
orders → stores	0
orders → staffs	0
order_items → orders	0
order_items → products	0
stocks → stores	0
stocks → products	0
staffs → stores	0
staffs → staffs (manager)	0

4. Phase 1 — Data Cleaning (Excel)

Raw CSV files were loaded into Microsoft Excel for preprocessing before being imported into MySQL. The following cleaning steps were applied to each table:

- Duplicate Detection: Used Excel's Remove Duplicates function on primary key columns for all 9 tables. No duplicates were found in any table.
- Null Value Handling:
 - customers.phone — allowed NULL (not all customers provided phone numbers)
 - customers.street, city, state, zip_code — allowed NULL (optional address fields)
 - orders.shipped_date — allowed NULL (unshipped orders have no shipped date)
 - staffs.manager_id — allowed NULL (top-level manager has no manager above them)
- String Standardization:
 - Applied TRIM() to all text columns to remove leading/trailing whitespace
 - Applied UPPER() to all state columns (NY, CA, TX)
 - Applied LOWER() to all email columns for consistency
 - Applied PROPER() to first_name and last_name columns
- Date Formatting: Verified all date columns (order_date, required_date, shipped_date) are in YYYY-MM-DD format. No format inconsistencies were found.

Business Logic Validation

- Confirmed required_date is always after or equal to order_date
- Confirmed shipped_date is always after order_date where not NULL
- Confirmed discount values are between 0 and 1 (0% to 100%)
- Confirmed list_price is always positive across products and order_items
- Confirmed model_year is within valid range (1900–2100)

8	Kali	Vargas	kali.vargas@bikes.shop	(972) 530-5555	1	3	1
9	Layla	Terrell	layla.terrell@bikes.shop	(972) 530-5556	1	3	7
10	Bernardine	Houston	bernardine.houston@bikes.shop	(972) 530-5557	1	3	7

4	Daryl	Spence	unknown	daryl.spence	988 Pearl Lar	Uniondale	NY	11553
5	Charolette	Rice	(916) 381-60	charolette.ri	107 River Dr.	Sacramento	CA	95820
6	Lyndsey	Bean	unknown	lyndsey.bean	769 West Ro	Fairport	NY	14450

5. Phase 2 — Database Creation & SQL Queries

Database Creation The retail_sales schema was created in MySQL Workbench using the EER Model Designer. Tables were defined with appropriate data types, NOT NULL constraints, primary keys, and foreign key relationships. The schema was forward engineered into a live MySQL 8.0.46 database.

Data Types Summary

Column Type	Data Type	Example Columns	Notes
Primary Keys	INT	brand_id, customer_id	Auto Integer
Names	VARCHAR(50–100)	first_name, product_name	-
Email	VARCHAR(100)	email	UNIQUE on staffs
State	CHAR(2)	state	Fixed 2-char
Zip Code	CHAR(5)	Zip_code	Fixed 5-char
Prices	DECIMAL(10,2)	list_price	-
Discount	DECIMAL(4,2)	discount	0.00 to 1.00
Dates	DATE	order_date, shipped_date	YYYY-MM-DD
Model Year	SMALLINT	model_year	e.g. 2016, 2018
Active Flag	SMALLINT	active (staffs)	1=active, 0=inactive

SQL Queries Six SQL scripts were written covering all required business analysis areas:

- Query File 1: Checking Data & Relationships
 - Verified row counts across all 9 tables and ran FK integrity checks to confirm zero orphaned rows across all relationships.

tbl	total
brands	9
categories	7
customers	1445
stores	3
staffs	10
products	321
orders	1615
order_items	4722
stocks	939

- Query File 2: Store-Wise & Region-Wise Analysis
 - Total revenue by store (store_name, city, state, total_orders, total_revenue)

- Revenue by state — region-wise aggregation
- Monthly sales trend broken down by store

store_name	city	state	total_orders	total_revenue
Baldwin Bikes	Baldwin	NY	1093	5215751.2775
Santa Cruz Bikes	Santa Cruz	CA	348	1605823.0365
Rowlett Bikes	Rowlett	TX	174	867542.2436

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- Query File 3: Product-Wise Sales & Inventory Trends
 - Top 10 best selling products by revenue
 - Sales aggregated by category
 - Sales aggregated by brand
 - Current inventory levels by product and store
 - Low stock alert — products with fewer than 5 units remaining

product_name	brand_name	category_name	units_sold	total_revenue
Trek Slash 8 27.5 - 2016	Trek	Mountain Bikes	154	555558.6111
Trek Conduit+ - 2016	Trek	Electric Bikes	145	389248.7025
Trek Fuel EX 8 29 - 2016	Trek	Mountain Bikes	143	368472.7294
Surly Straggler 650b - 2016	Surly	Cyclocross Bicycles	151	226765.5510
Trek Domane SLR 6 Disc - 2017	Trek	Road Bikes	43	211584.6153
Surly Straggler - 2016	Surly	Cyclocross Bicycles	147	203507.6200
Trek Remedy 29 Carbon Frameset - 2016	Trek	Mountain Bikes	125	203380.8701
Trek Powerfly 8 FS Plus - 2017	Trek	Electric Bikes	41	188249.6235
Trek Madone 9.2 - 2017	Trek	Road Bikes	39	175899.6482
Trek Silque SLR 8 Women's - 2017	Trek	Road Bikes	29	174524.7315

- Query File 4: Customer Orders & Order Frequency
 - Total orders and spend per customer
 - Repeat customers — customers with more than 1 order
 - Top 10 highest spenders
 - Customers who have never placed an order

customer_name	email	total_spent
Sharyn Hopkins	sharyn.hopkins@hotmail.com	34807.9392
Pamelia Newman	pamelia.newman@gmail.com	33634.2604
Abby Gamble	abby.gamble@aol.com	32803.0062
Lyndsey Bean	lyndsey.bean@hotmail.com	32675.0725
Emmitt Sanchez	emmitt.sanchez@hotmail.com	31925.8857
Melanie Hayes	melanie.hayes@msn.com	31913.6902
Debra Burks	debra.burks@yahoo.com	27888.1834
Elinore Aguilar	elinore.aguilar@msn.com	25636.4531
Corrina Sawyer	corrina.sawyer@yahoo.com	25612.7021
Shena Carter	shena.carter@yahoo.com	24890.6244

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- Query File 5: Staff Performance Reports
 - Revenue generated per staff member with store breakdown
 - Staff order count by month
 - Manager and team performance — hierarchical revenue breakdown

staff_name	store_name	total_orders	total_revenue
Marcelene Boyer	Baldwin Bikes	553	2624120.6530
Venita Daniel	Baldwin Bikes	540	2591630.6245
Genna Serrano	Santa Cruz Bikes	184	853287.3589
Mireya Copeland	Santa Cruz Bikes	164	752535.6776
Kali Vargas	Rowlett Bikes	88	463918.3046
Layla Terrell	Rowlett Bikes	86	403623.9390

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- Query File 6: Revenue & Discount Analysis
 - Overall gross revenue, total discount, and net revenue summary
 - Revenue and discount breakdown by year
 - Average discount percentage by product category
 - Orders with high discounts — all orders where discount exceeds 20%

gross_revenue	total_discount	net_revenue
8578988.88	889872.3224	7689116.5576

6. Phase 3 — SQL Views

Six SQL Views were created to provide reusable, pre-aggregated datasets for Tableau. Each view encapsulates complex JOIN and aggregation logic, allowing Tableau to connect directly without requiring additional transformation.

View Name	Primary Use	Key Columns
vw_store_sales	Dashboard 1 — Sales	store_name, city, state, yr_month, total_orders, net_revenue
vw_product_sales	Dashboard 1 — Sales	product_name, brand_name, category_name, store_name, yr_month, units_sold, net_revenue
vw_inventory	Dashboard 2 — Inventory	store_name, product_name, brand_name, stock_remaining, stock_status
vw_staff_performance	Dashboard 3 — Staff	staff_name, manager_name, store_name, yr_month, total_orders, net_revenue
vw_customer_analysis	Dashboard 3 — Customer	customer_name, email, city, state, total_orders, net_spent, customer_type
vw_revenue_discount	Dashboard 2 — Discount	order_date, yr_month, store_name, product_name, category_name, gross_amount, discount_amount, net_amount

Stock Status Classification

The vw_inventory view includes a derived stock_status column using a CASE expression:

- WHEN quantity = 0 THEN 'Out of Stock'
- WHEN quantity < 5 THEN 'Low Stock'
- ELSE 'In Stock'

Customer Segmentation Logic

The vw_customer_analysis view classifies customers into three segments:

- One Time — customers with exactly 1 order
- Returning — customers with 2 to 5 orders
- Loyal — customers with 6 or more orders

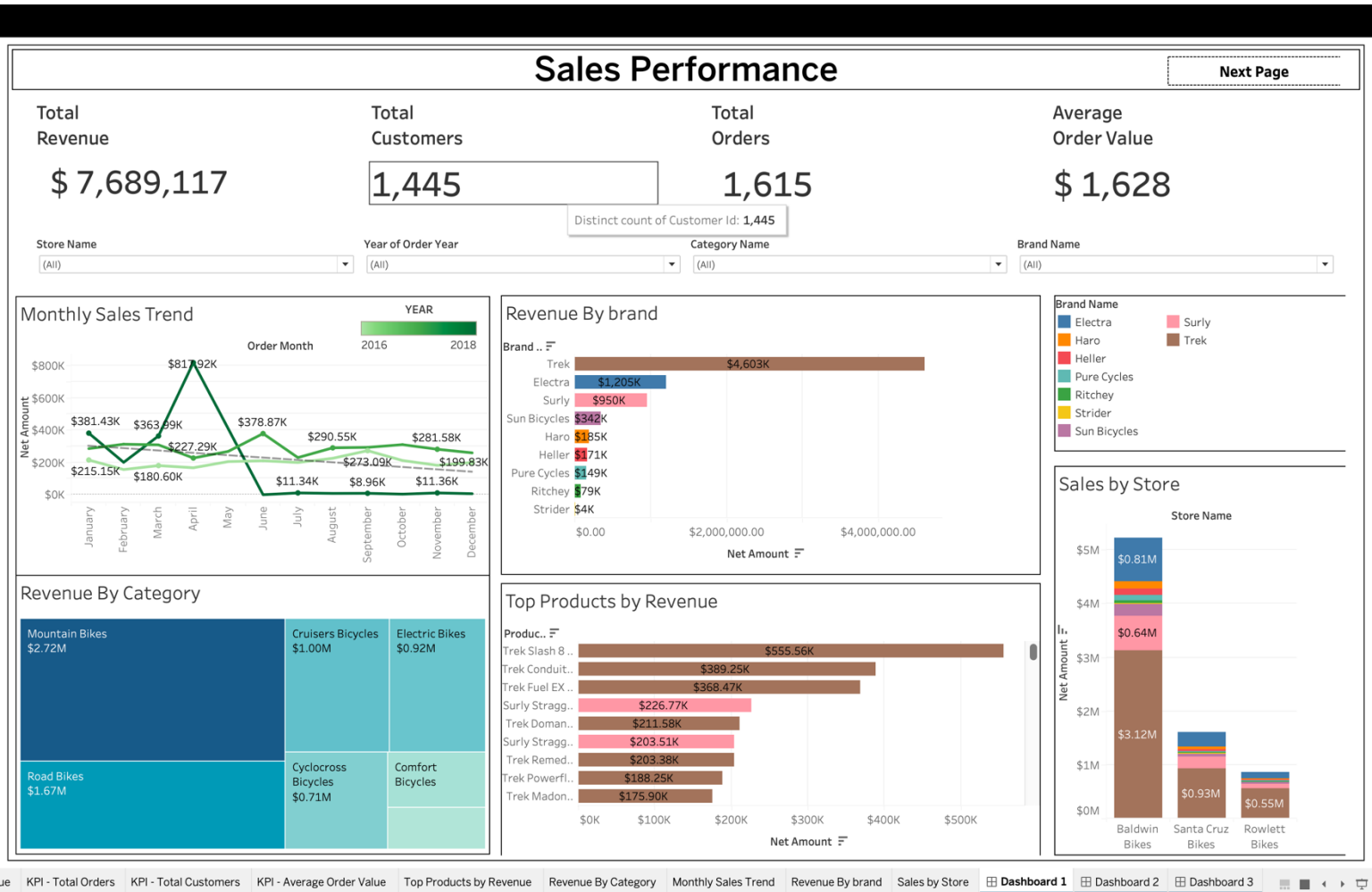
Tables_in_retail_sales	Table_type	
vw_customer_analysis	VIEW	
vw_inventory	VIEW	
vw_product_sales	VIEW	
vw_revenue_discount	VIEW	
vw_staff_performance	VIEW	
vw_store_sales	VIEW	

7. Phase 4 — Tableau Dashboards

Three interactive dashboards were built in Tableau Desktop using the six SQL Views exported as CSVs. Each dashboard includes KPI cards, multiple charts, and interactive filters. Navigation buttons (Previous Page / Next Page) allow seamless movement between dashboards.

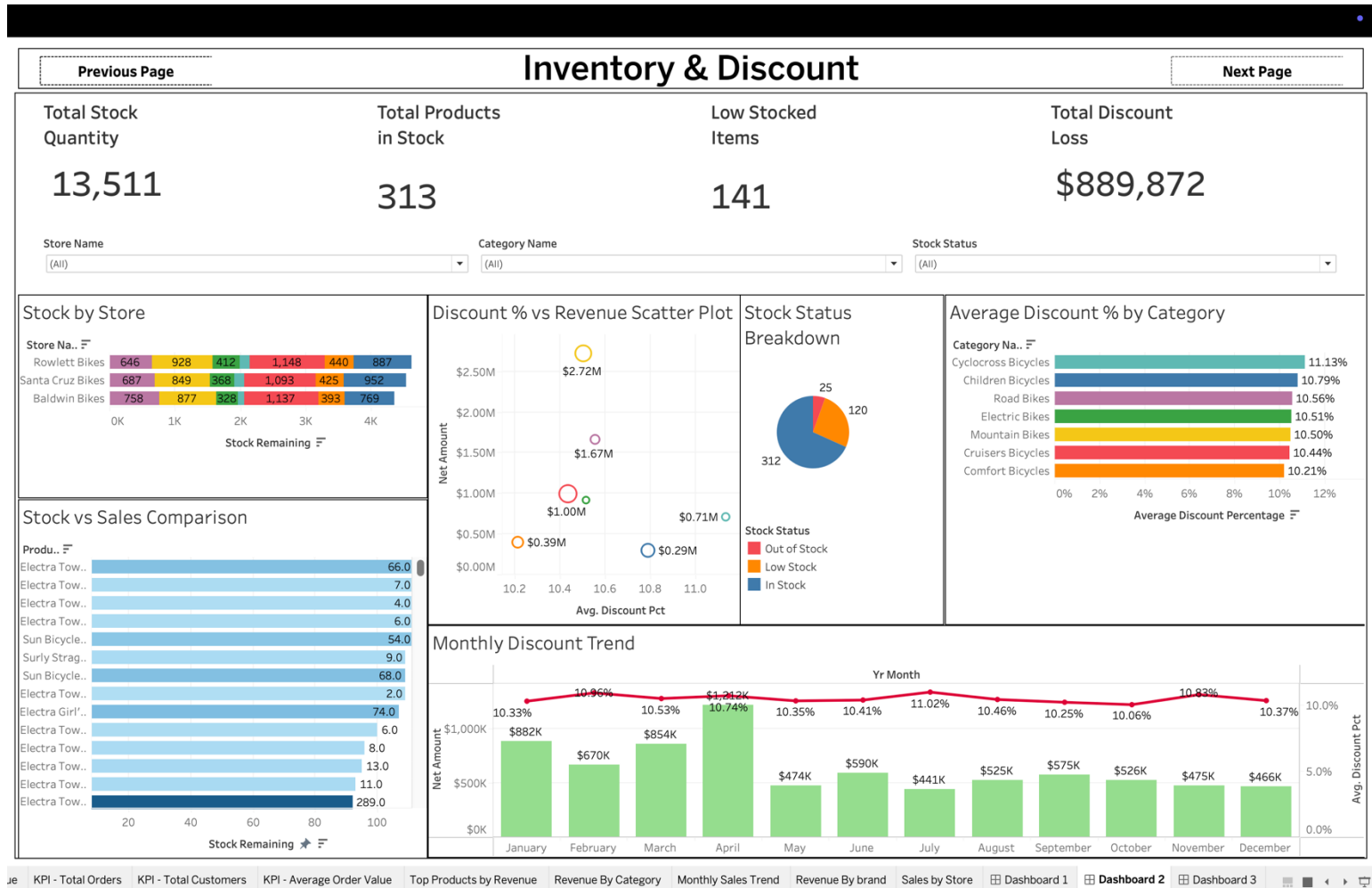
Dashboard 1 — Sales Performance

- Data Source: vw_revenue_discount, vw_product_sales, vw_store_sales
- KPI Cards: Total Revenue: \$7,689,117 | Total Customers: 1,445 | Total Orders: 1,615 | Average Order Value: \$1,628
- Charts:
 - Monthly Sales Trend — line chart by year showing revenue across all 12 months
 - Revenue by Brand — horizontal bar chart ranked by net revenue
 - Revenue by Category — treemap sized and colored by revenue
 - Top 10 Products by Revenue — horizontal bar chart with top filter
 - Sales by Store — stacked bar chart showing revenue per store colored by brand
- Filters: Store Name, Year of Order, Category Name, Brand Name



Dashboard 2 — Inventory & Discount

- Data Source: vw_inventory, vw_revenue_discount
- KPI Cards: Total Stock Quantity: 13,511 | Total Products in Stock: 313 | Low Stocked Items: 141 | Total Discount Loss: \$889,872
- Charts:
 - Stock by Store — horizontal bar showing total inventory per store by category
 - Discount % vs Revenue Scatter Plot — bubbles per category showing discount vs net revenue
 - Stock Status Breakdown — donut chart (In Stock / Low Stock / Out of Stock)
 - Average Discount % by Category — horizontal bar ranked by discount
 - Stock vs Sales Comparison — bar chart of stock remaining per product
 - Monthly Discount Trend — dual axis chart: revenue bars + discount % line
- Filters: Store Name, Category Name, Stock Status



Monthly Discount Trend

Yr Month

10.33%

10.96%

10.53%

10.74%

10.35%

10.41%

11.02%

10.46%

10.25%

10.06%

10.83%

10.37%

Net Amount

\$882K

\$670K

\$854K

\$1,212K

\$474K

\$590K

\$441K

\$525K

\$575K

\$526K

\$475K

\$466K

January

February

March

April

May

June

July

August

September

October

November

December

0%

5%

10%

Net Amount

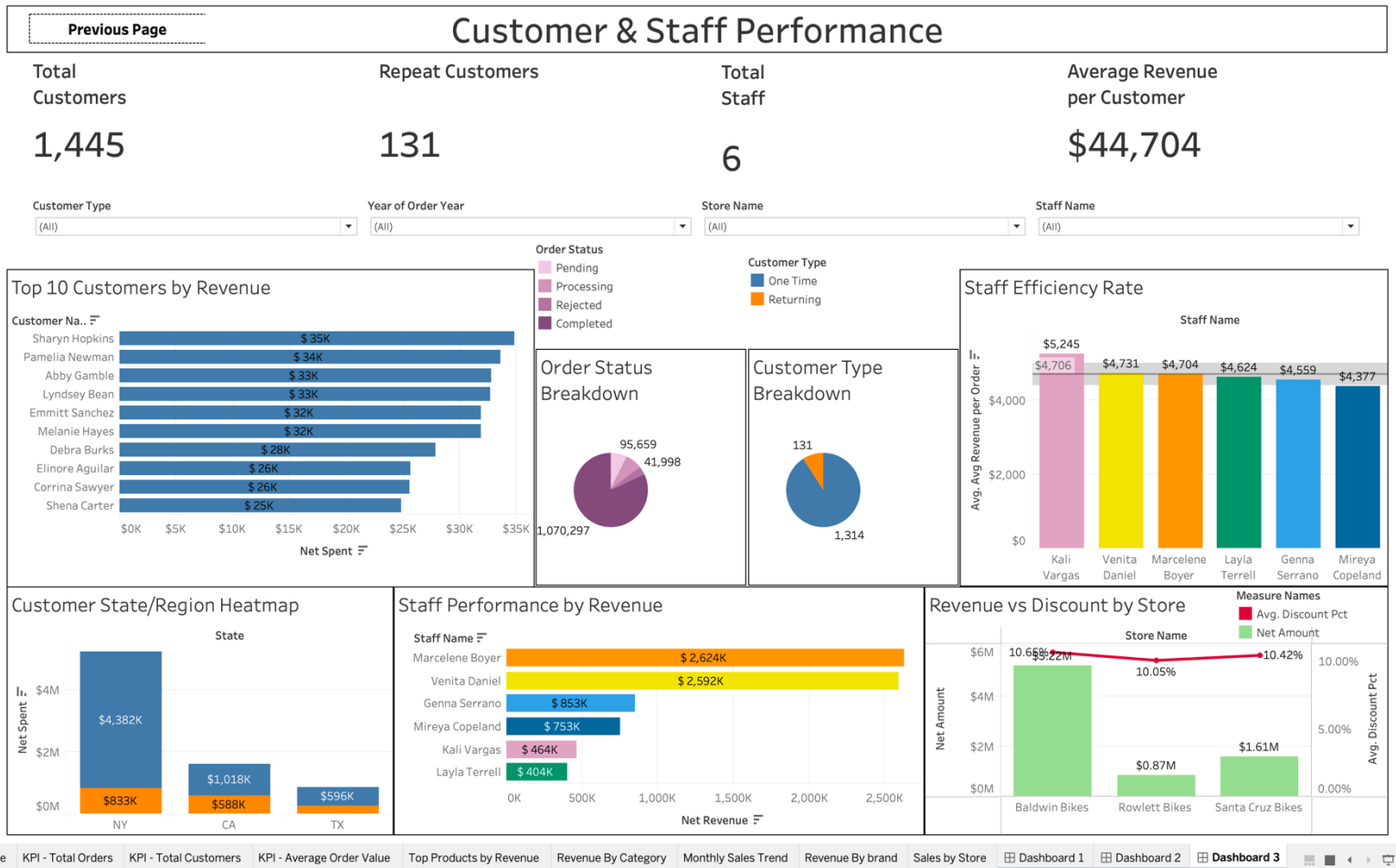
Avg. Discount Pct

je KPI - Total Orders KPI - Total Customers KPI - Average Order Value Top Products by Revenue Revenue By Category Monthly Sales Trend Revenue By brand Sales by Store Dashboard 1 Dashboard 2 Dashboard 3

Dashboard 3 — Customer & Staff Performance

- Data Sources: vw_customer_analysis, vw_staff_performance
- KPI Cards: Total Customers: 1,445 | Repeat Customers: 131 | Total Staff: 6 (active staff with orders) | Average Revenue per Customer: \$44,704
- Charts:
 - Top 10 Customers by Revenue — horizontal bar sorted by net_spent
 - Order Status Breakdown — donut chart (Pending / Processing / Rejected / Completed)
 - Customer Type Breakdown — donut chart (One Time / Returning)
 - Staff Efficiency Rate — bar chart showing avg revenue per order per staff member
 - Customer State/Region — bar chart showing net spend by state (NY, CA, TX)
 - Staff Performance by Revenue — horizontal bar per staff with color coding
 - Revenue vs Discount by Store — dual axis bar + line chart
- Filters: Customer Type, Year of Order, Store Name, Staff Name

Customer & Staff Performance



8. Business Insights

Sales Performance

- Baldwin Bikes is the top performing store, generating \$3.12M — approximately 65% of total net revenue, far ahead of Santa Cruz Bikes (\$0.93M) and Rowlett Bikes (\$0.55M).
- Trek dominates brand sales at \$4.6M, nearly 4x the second-highest brand Electra at \$1.2M.
- Every top 10 product by revenue is a Trek or Surly product.
- Trek Slash 8 is the single best-selling product at \$555K, followed by Trek Conduit at \$389K and Trek Fuel EX at \$368K.
- Mountain Bikes is the top revenue category at \$2.72M, followed by Road Bikes at \$1.67M.
- Children Bicycles and Comfort Bicycles are the weakest performers.
- Sales peaked in April across all years at \$817K and dropped sharply through summer, with revenue falling as low as \$8.96K in July — a clear seasonal pattern.

Inventory & Discount

- 141 out of 313 products (45%) are in Low Stock status, and 25 products are completely Out of Stock — representing direct lost sales.
- Total discount loss is \$889,872 — nearly 11.6% of gross revenue is being given away in discounts with no strategic differentiation across categories.
- Average discount % is remarkably flat across all categories (10.21% to 11.13%) suggesting blanket discounting rather than targeted promotions.

- Monthly discount % remains constant at ~10-11% even during summer revenue drops, meaning discounts are not being used to counter seasonal slowdowns.
- Baldwin Bikes applies the highest average discount at 10.65% despite already being the top-revenue store.

Customer & Staff Performance

- 1,314 out of 1,445 customers (91%) are one-time buyers — the business is almost entirely dependent on new customer acquisition rather than retention.
- Only 131 customers are repeat buyers, and zero customers qualify as Loyal (6+ orders) under current segmentation thresholds.
- Sharyn Hopkins is the top customer at \$35K lifetime spend, followed by Pamela Newman at \$34K.
- New York generates \$4.38M in customer spend, far ahead of California at \$1.01M and Texas at \$0.60M — heavily concentrated in one region.
- Marcelene Boyer and Venita Daniel are the top two performing staff, each generating ~\$2.6M, significantly ahead of the remaining four staff combined.
- Kali Vargas achieves the highest staff efficiency rate at \$5,245 average revenue per order, indicating strong upselling capability despite lower order volume.

9. Business Recommendations

Store Strategy

- Invest further in Baldwin Bikes — it generates 65% of revenue, making it the highest-ROI location for marketing spend, expanded inventory, and additional staffing.
- Conduct a root-cause analysis for Rowlett Bikes — at \$0.55M it generates less than 18% of Baldwin's revenue despite similar operating infrastructure. Investigate product mix, staffing quality, and local market demand.
- Develop a Santa Cruz growth plan — at \$0.93M it has potential but needs targeted interventions in inventory assortment and customer outreach to close the gap with Baldwin.

Inventory Management

- Immediately restock the 25 out-of-stock products — prioritize Trek and Mountain Bike category items as they are the highest revenue drivers.
- Cross-reference the 141 low-stock products with sales velocity data to distinguish urgently needed restocks from slow-moving items that don't require replenishment.
- Run targeted clearance promotions on overstocked slow-moving items (e.g. Electra Townie Original at 289 units) to free up capital and floor space.

Discount Strategy

- Replace blanket 10-11% discounts with a targeted discount framework — apply higher discounts only to slow-moving products and categories, and reserve full-price sales for high-demand Trek and Mountain Bike products.
- Deploy summer discount promotions to counter the seasonal revenue drop — revenue falls by over 95% between April and July, yet discount rates remain unchanged.
- Review Baldwin Bikes' discount policy — as the top store it should not need to offer the highest discounts. Reducing discounts at Baldwin could recover significant margin.

Customer Retention

- A 91% one-time buyer rate is a critical business risk — launch a customer retention program including post-purchase follow-up emails, loyalty points, and repeat-purchase incentives.

- Identify and target the top 50 customers by lifetime spend with VIP treatment — early access to new models, dedicated service, and personalized promotions.
- Invest in growing the CA and TX customer base — these markets currently account for only 26% of revenue combined despite housing two stores.

Staff Development

- Study Marcelene Boyer and Venita Daniel's sales approach and develop internal training programs to bring lower-performing staff closer to their output.
- Leverage Kali Vargas's high-value order closing skills — her average revenue per order is the highest on the team and suggests strong consultative selling capability.
- Investigate Layla Terrell's low performance at \$404K — determine whether it is a training, territory, or motivation issue and address it with a structured improvement plan.

Product Strategy

- Expand the Trek and Surly product catalogue — the top 10 revenue products are almost exclusively from these two brands, making expansion low-risk and high-reward.
- Review underperforming brands — Ritchey (\$79K), Pure Cycles (\$149K), and Heller (\$171K) contribute minimally to revenue and may be candidates for SKU reduction.
- Allocate more shelf space, marketing budget, and staff training toward Mountain Bikes — as the top category at \$2.72M, it deserves disproportionate investment.

10. Conclusion

This project successfully designed and implemented a comprehensive Retail Sales and Inventory Intelligence System for a multi-store bicycle retail company. By following a structured analytics pipeline — from data cleaning in Excel, to relational database design in MySQL, to interactive visualization in Tableau — the project delivers actionable intelligence across all key dimensions of retail operations.

The analysis reveals a business with strong top-line revenue (\$7.69M) but significant opportunities for improvement in three critical areas: inventory management (45% of products at low stock), customer retention (91% one-time buyer rate), and discount strategy (nearly \$900K in undifferentiated discount spend annually).

The Tableau dashboards provide an accessible, interactive interface for business stakeholders to monitor KPIs, explore trends, and identify issues in real time — reducing dependence on manual reporting and enabling faster, data-driven decision making. Implementing the recommendations outlined in this report — particularly around customer retention, targeted discounting, and inventory restocking — has the potential to meaningfully improve both revenue and profitability in the next operating period.

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